CHE 311 Homework Week 10

1. Using spherical funnel to measure the viscosity of wine. The funnel was fully filled up initially. The draining time (t_{efflux}) was measured 100 seconds. The radius of the tank is 5 cm. The radius of the pipe is 1 mm and the length is 10 cm. The density of the wine in room temperature is about 1000 kg/m^3. Please calculate the viscosity of the wine in this case.

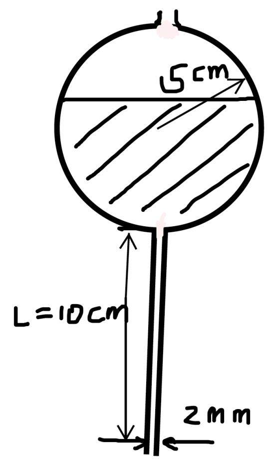

Figure 1. Dimensions of the spherical funnel.

2. The 45^o conical funnel (as in Figure 2) was filled up with water at room temperature. The radius of the funnel at the open surface is 5 cm. The length of the vertical draining tube is 10 cm and its radius is 1mm. How long does it take to completely drain the funnel?

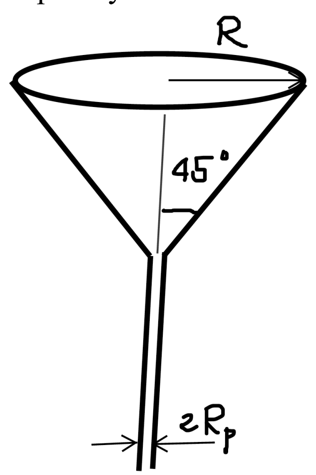

Figure 2. Conical funnel.

3. BSL, P228, 7B.9
4. BSL, P228, 7B.10 (a) – (c)